Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			(calculating the percentage change) allows the results to be comparable (1) Initial masses of worm would differ (1)		2		2		2
	(b)	(i)		Linear scale (1) including figure at origin and no break line Correct plots (1) ± ½ small square Line drawn (with no extrapolation) (1) extrapolation is acceptable with a curve of best fit		3		3	3	3
		(ii)		**Any four (x1) from:** 1. Increase to <u>60</u> and then levels off(1) 2. (the body fluids of the marine) worms {are hypertonic to /have a low<u>er</u> water potential than} {the dilute seawater/ the solution}/ ORA(1) 3. The worms gain water by <u>osmosis</u> (resulting in an increase in mass)(1) 4. (during the first 15 mins) the increase in mass is greater due to the steepness of the water potential gradient(1) 5. (after 60 mins there is no change in mass as) equilibrium is reached / isotonic/ no net movement of water/ OWTTE (1)		4		4		3
	(c)			As a control (1) to show that any change in mass was due to the worms being placed in diluted seawater (and no other variable).(1)			2	2		2
	(d)	(i)		The percentage change in mass of *Golfingia* {returns to zero/ decreases} (after 45 mins) whereas the mass of *Nereis* {continued to increase/ plateaus} (1) The maximum percentage increase in mass in *Golfingia* {is much lower / is reached sooner} than that of *Nereis/* equilibrium is reached quicker in *Golfingia* than in *Nereis*(1)		2		2		2